**Introduction to CNN**

Yann LeCun, director of Facebook's AI Research Group, pioneered convolutional neural networks. In 1988, he built the first one, LeNet, which was used for character recognition tasks like reading zip codes and digits.

Have you ever wondered how facial recognition works on social media, or how object detection helps in building self-driving cars, or how disease detection is done using visual imagery in healthcare? It's all possible thanks to convolutional neural networks (CNN). Here's an example of convolutional neural networks that illustrates how they work:

Imagine there's an image of a bird, and you want to identify whether it's really a bird or some other object. The first thing you do is feed the pixels of the image in the form of arrays to the input layer of the neural network (multi-layer networks used to classify things). The hidden layers carry out feature extraction by performing different calculations and manipulations. There are multiple hidden layers like the convolution layer, the ReLU layer, and pooling layer, that perform feature extraction from the image. Finally, there's a fully connected layer that identifies the object in the image.

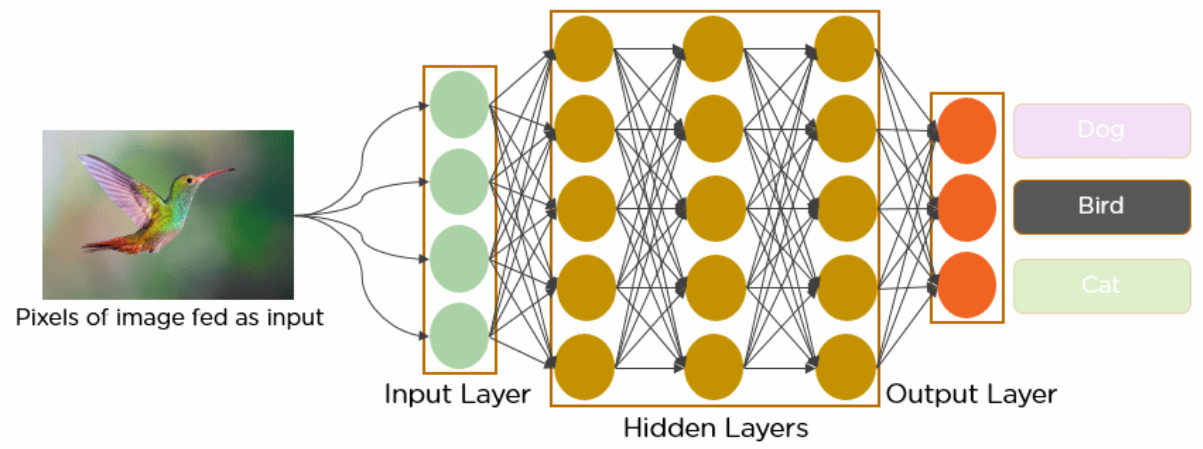
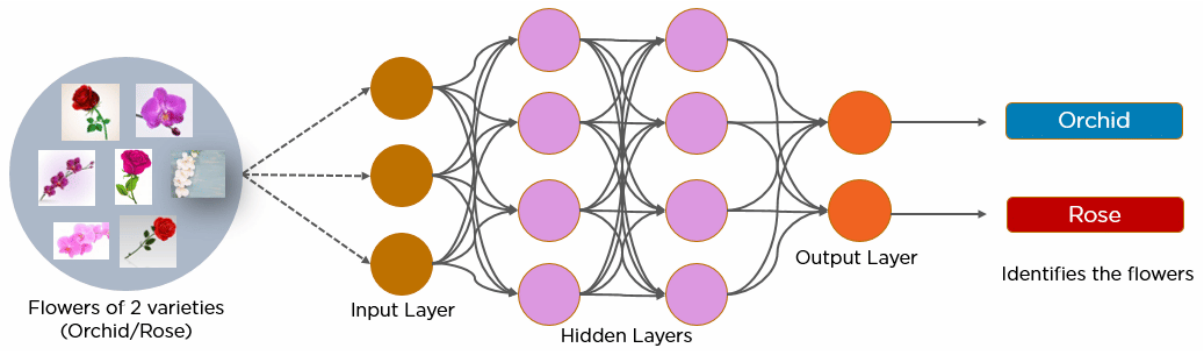



Fig: Convolutional Neural Network to identify the image of a bird

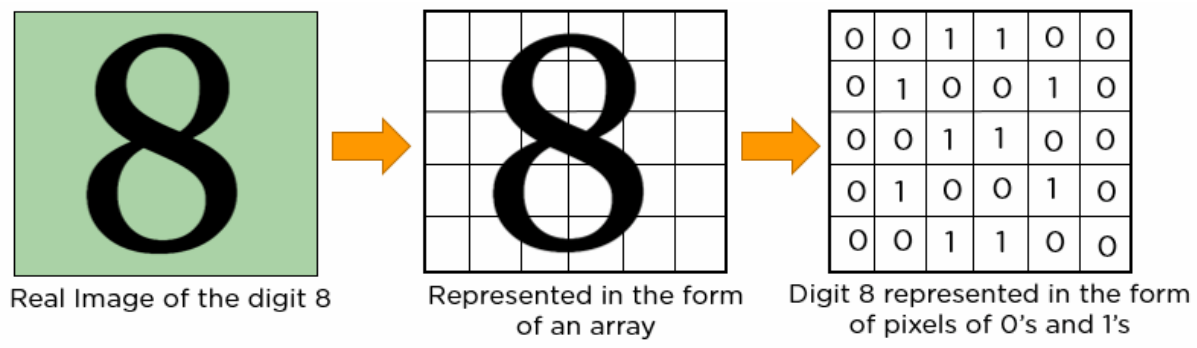
**What is Convolutional Neural Network?**

A convolutional neural network is a feed-forward neural network that is generally used to analyze visual images by processing data with grid-like topology. It's also known as a ConvNet. A convolutional neural network is used to detect and classify objects in an image.

Below is a neural network that identifies two types of flowers: Orchid and Rose.




In CNN, every image is represented in the form of an array of pixel values.



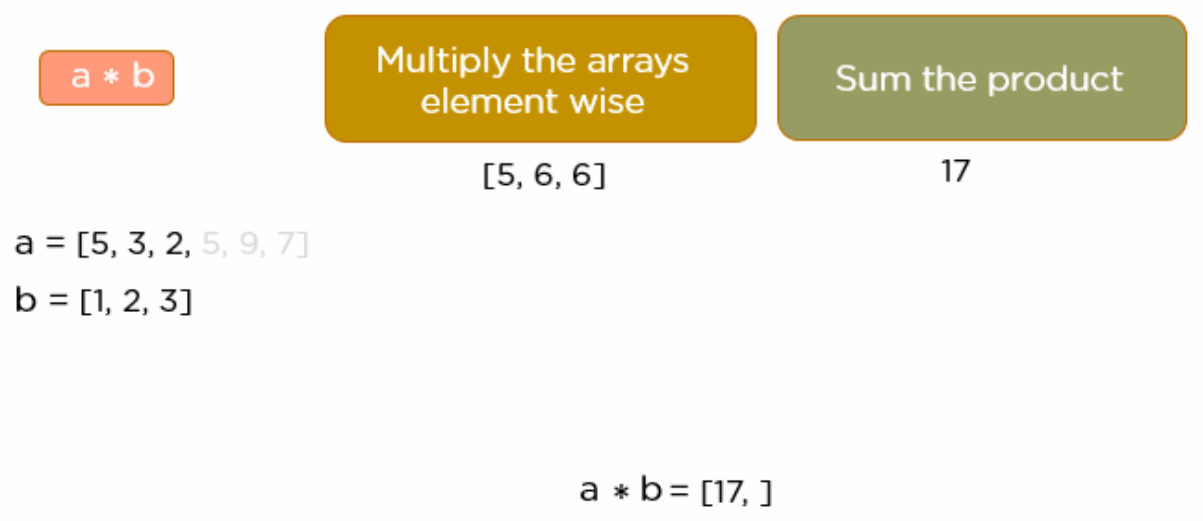

The convolution operation forms the basis of any convolutional neural network. Let's understand the convolution operation using two matrices, a and b, of 1 dimension.

a = [5,3,7,5,9,7]

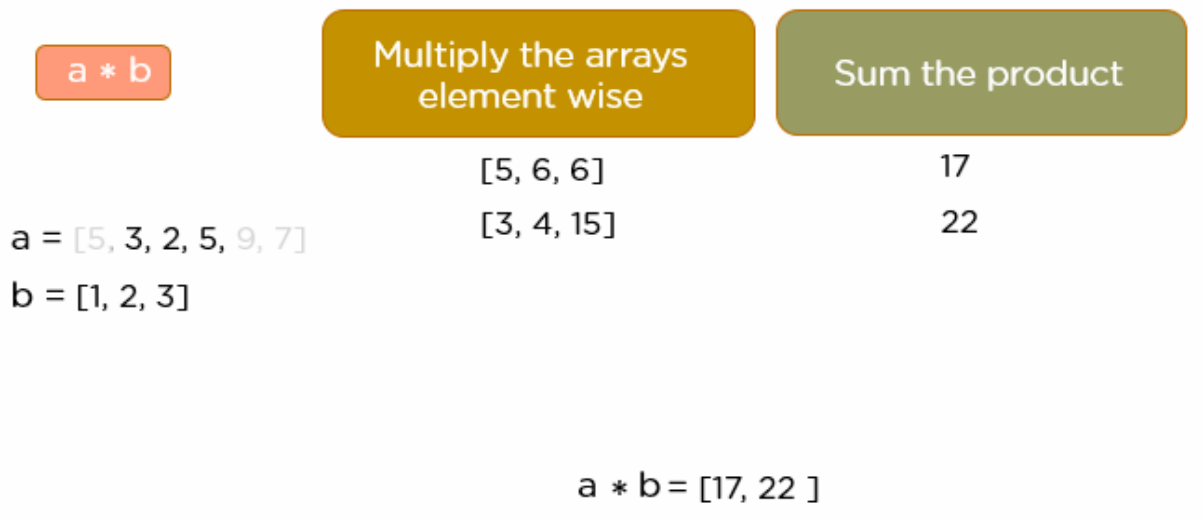
b = [1,2,3]

In convolution operation, the arrays are multiplied element-wise, and the product is summed to create a new array, which represents a*b.

The first three elements of the matrix a are multiplied with the elements of matrix b. The product is summed to get the result.



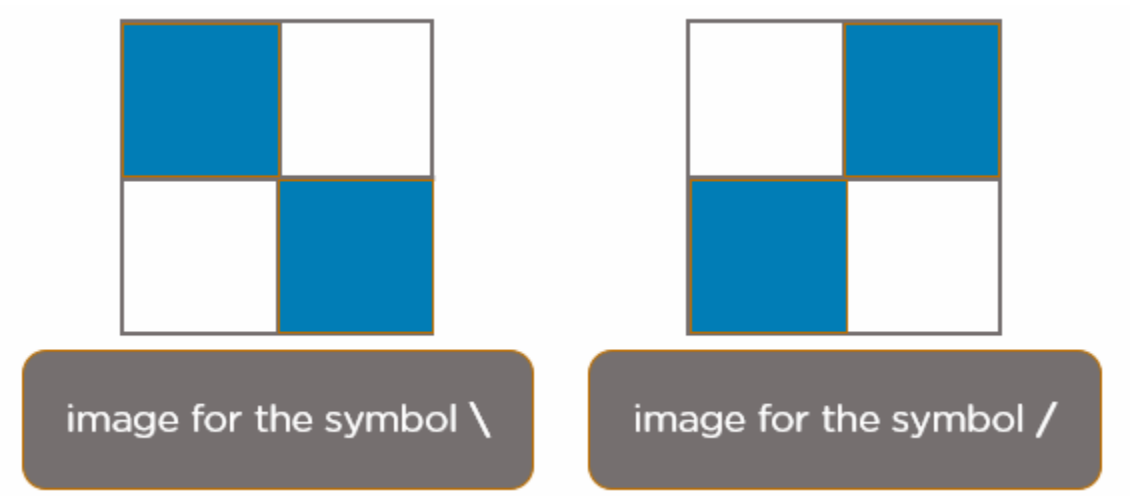

The next three elements from the matrix a are multiplied by the elements in matrix b, and the product is summed up.




This process continues until the convolution operation is complete.

**How Does CNN Recognize Images?**

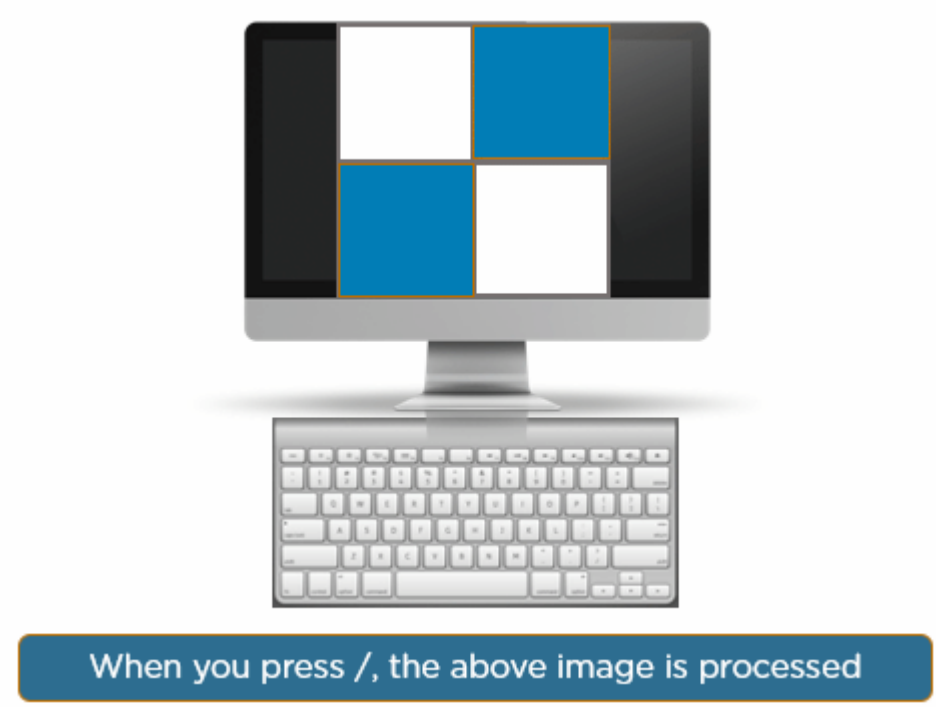
Consider the following images:



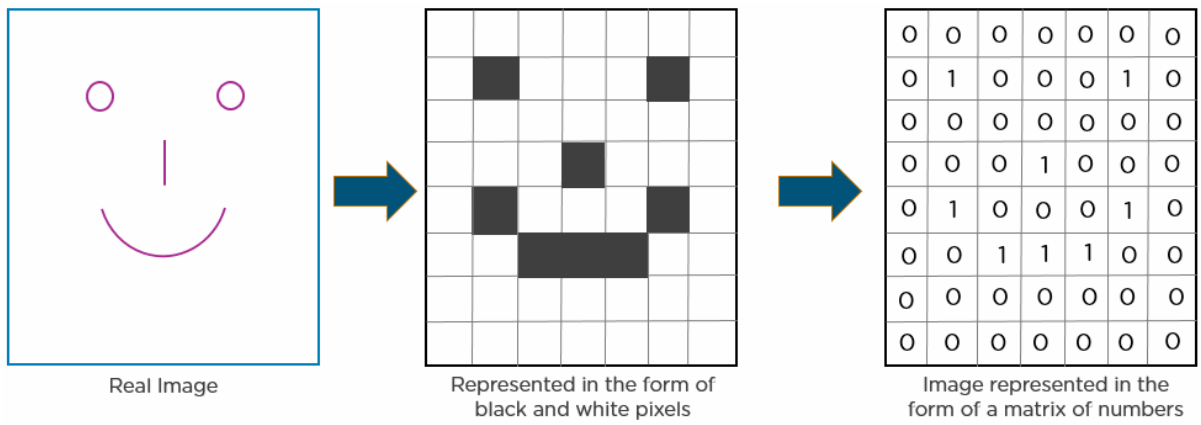

The boxes that are colored represent a pixel value of 1, and 0 if not colored.

When you press backslash (\), the below image gets processed.


When you press forward-slash (/), the below image is processed:




Here is another example to depict how CNN recognizes an image:




As you can see from the above diagram, only those values are lit that have a value of 1.



**Layers in a Convolutional Neural Network**

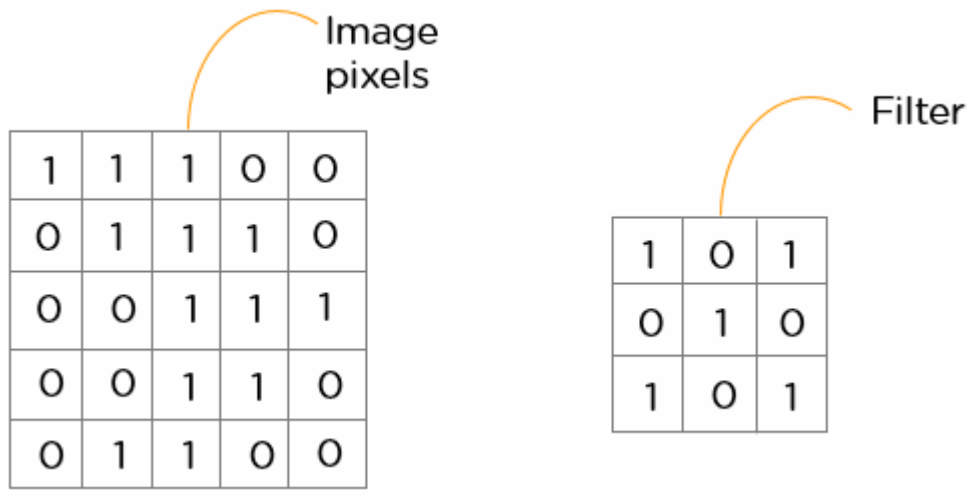
A convolution neural network has multiple hidden layers that help in extracting information from an image. The four important layers in CNN are:

1. Convolution layer
2. ReLU layer
3. Pooling layer
4. Fully connected layer
5. ReLU layer/ Activation Layer
6. Flattening
7. Output Layer

**Convolution Layer**

This is the first step in the process of extracting valuable features from an image. A convolution layer has several filters that perform the convolution operation. Every image is considered as a matrix of pixel values.

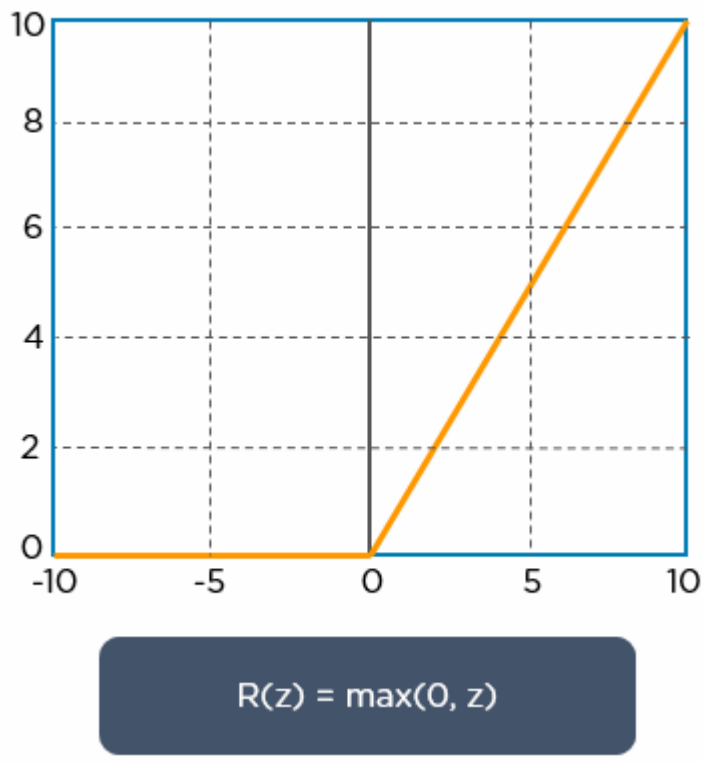
Consider the following 5x5 image whose pixel values are either 0 or 1. There’s also a filter matrix with a dimension of 3x3. Slide the filter matrix over the image and compute the dot product to get the convolved feature matrix.



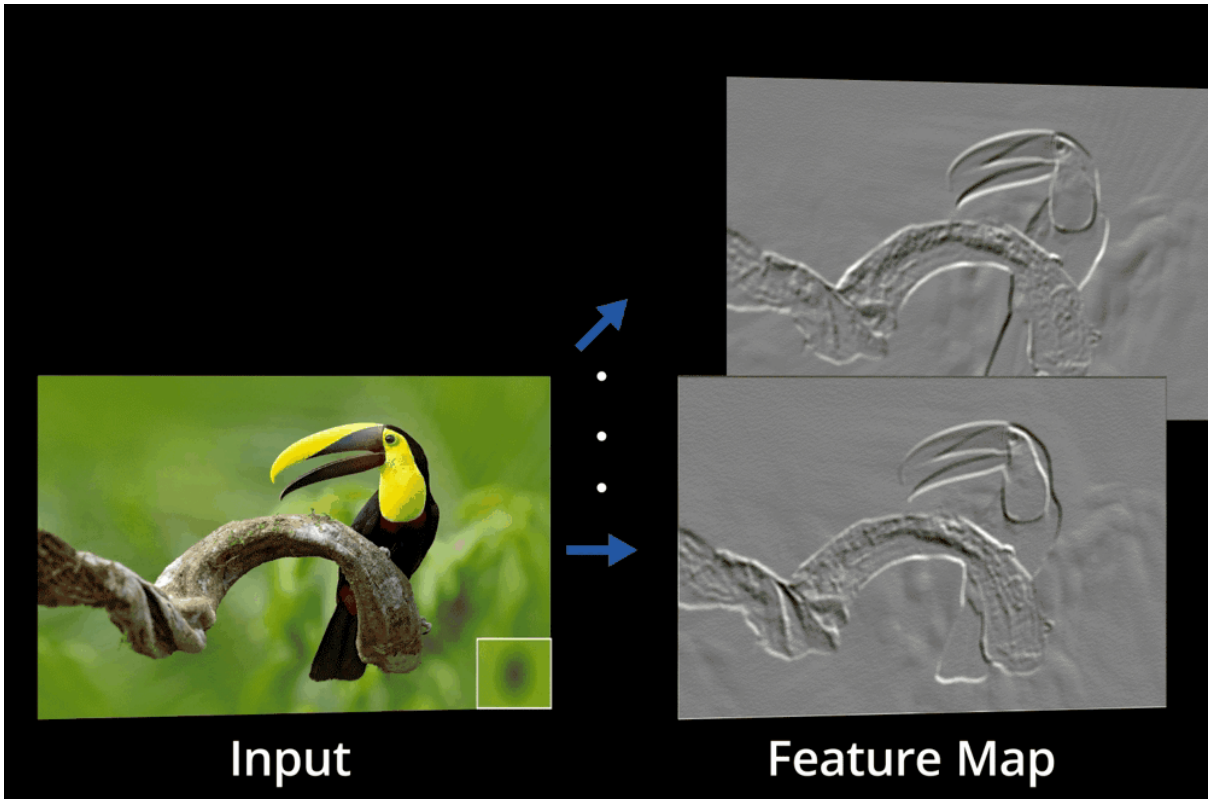

**ReLU layer**

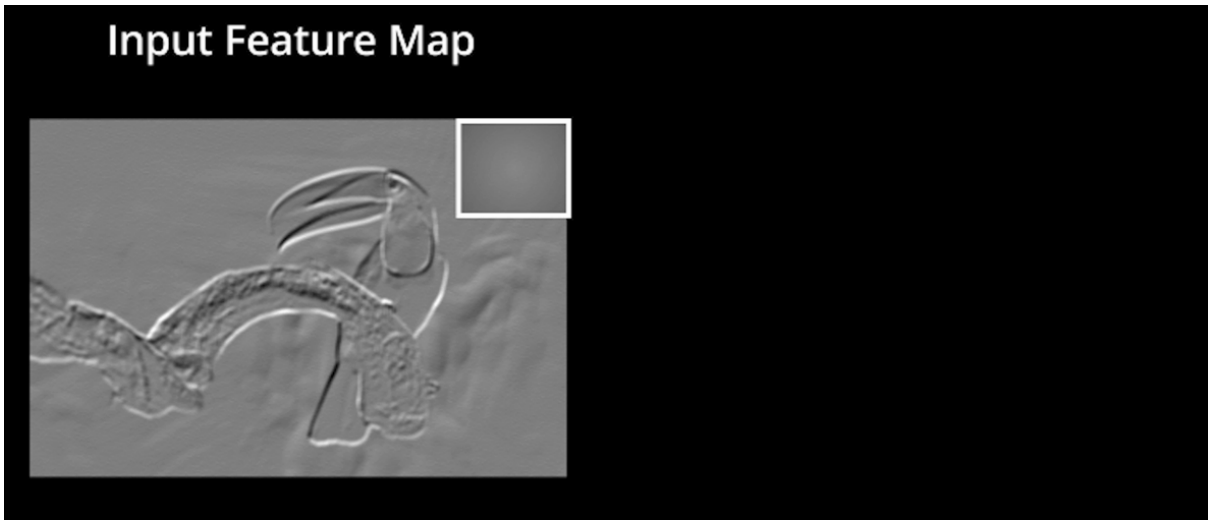
ReLU stands for the rectified linear unit. Once the feature maps are extracted, the next step is to move them to a ReLU layer.

ReLU performs an element-wise operation and sets all the negative pixels to 0. It introduces non-linearity to the network, and the generated output is a rectified feature map. Below is the graph of a ReLU function:



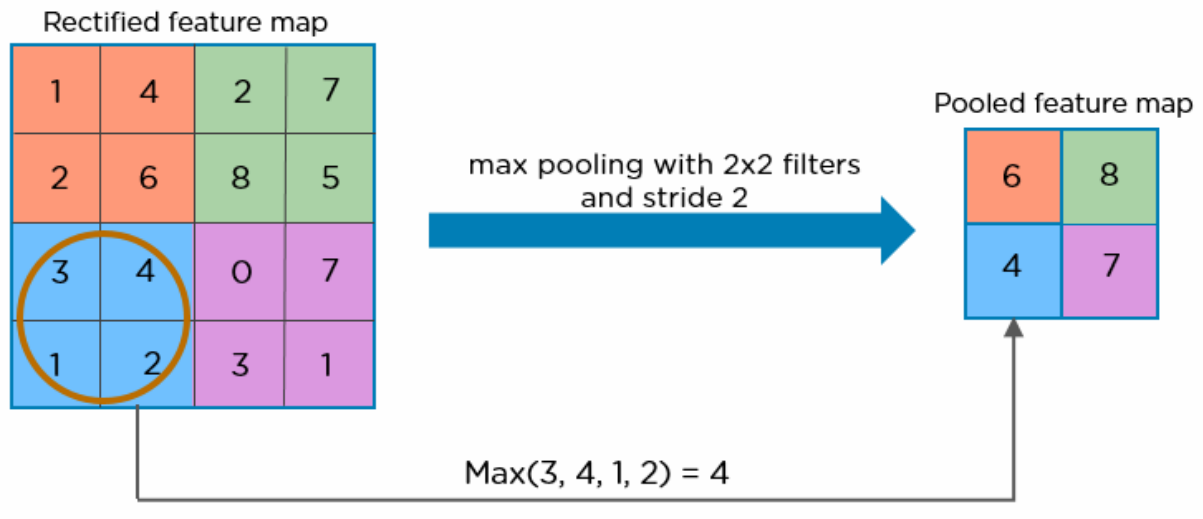

The original image is scanned with multiple convolutions and ReLU layers for locating the features.



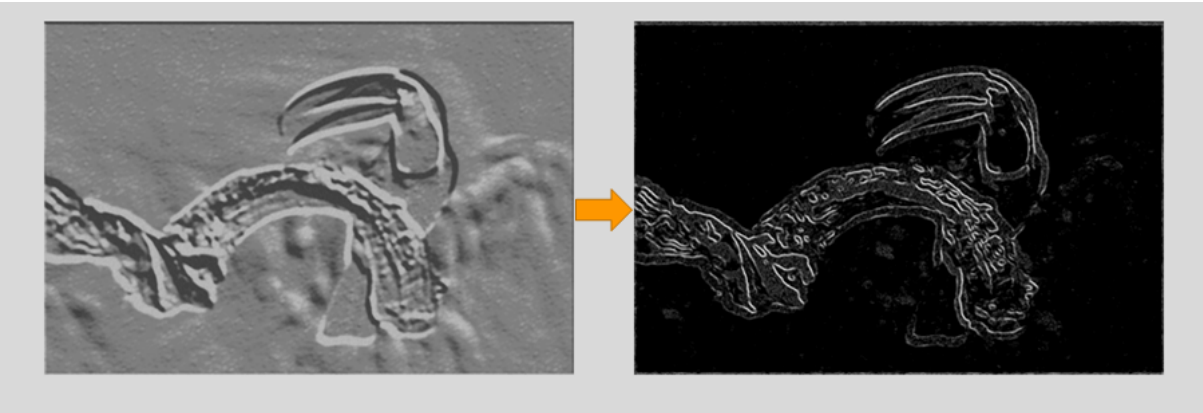




**Pooling Layer**

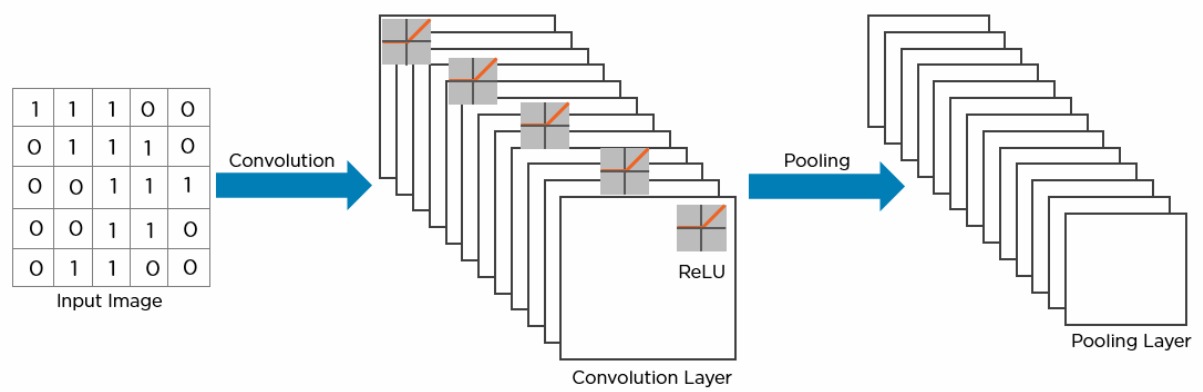
Pooling is a down-sampling operation that reduces the dimensionality of the feature map. The rectified feature map now goes through a pooling layer to generate a pooled feature map.



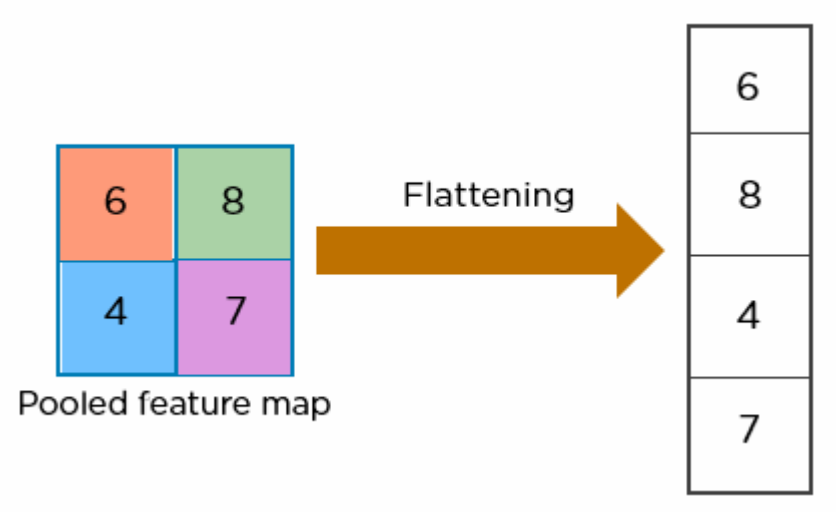

The pooling layer uses various filters to identify different parts of the image like edges, corners, body, feathers, eyes, and beak.



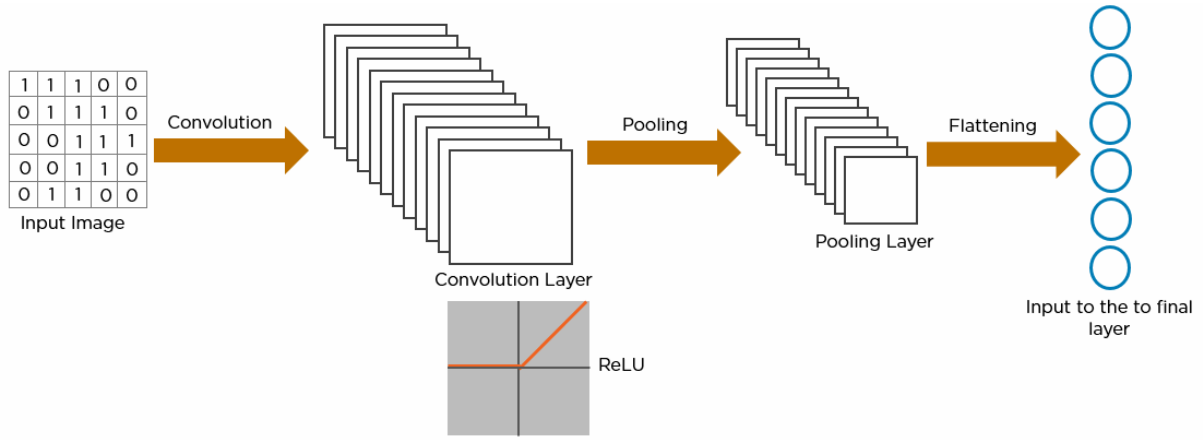
Here's how the structure of the convolution neural network looks so far:

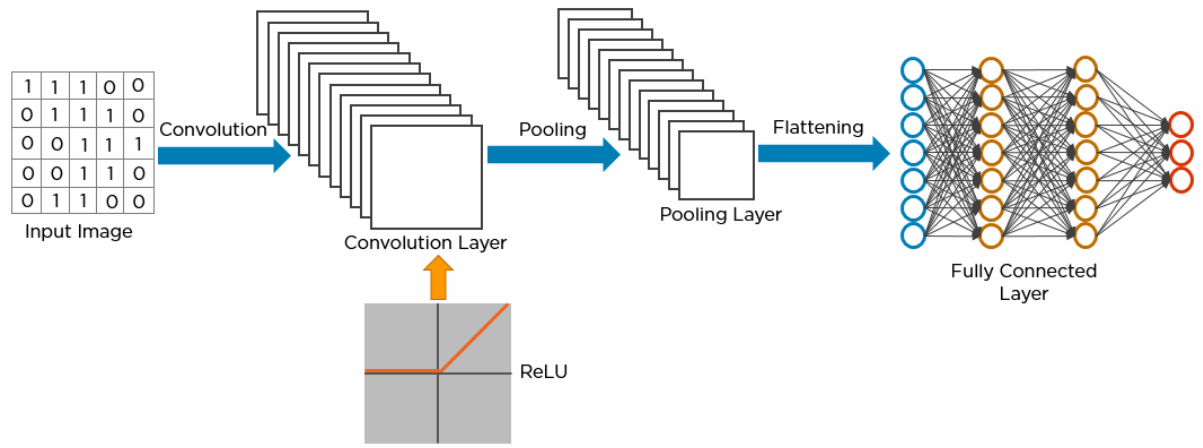


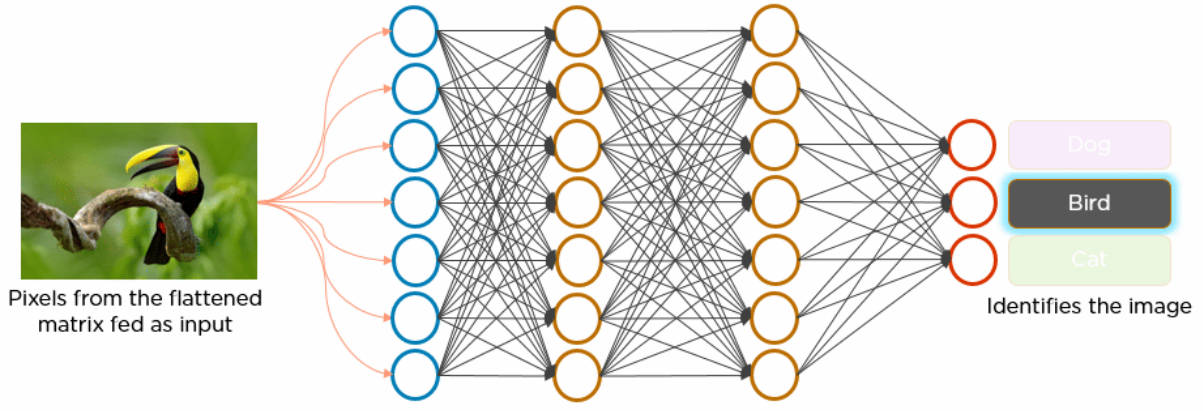

The next step in the process is called flattening. Flattening is used to convert all the resultant 2-Dimensional arrays from pooled feature maps into a single long continuous linear vector.




The flattened matrix is fed as input to the fully connected layer to classify the image.

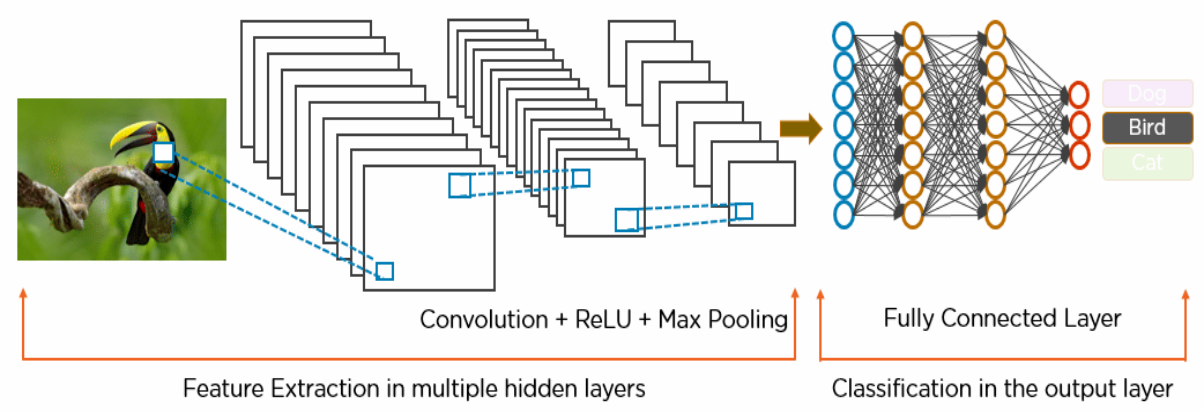





Here's how exactly CNN recognizes a bird:

- The pixels from the image are fed to the convolutional layer that performs the convolution operation
- It results in a convolved map
- The convolved map is applied to a ReLU function to generate a rectified feature map
- The image is processed with multiple convolutions and ReLU layers for locating the features
- Different pooling layers with various filters are used to identify specific parts of the image
- The pooled feature map is flattened and fed to a fully connected layer to get the final output




- **Activation Layer**

The activation layer introduces nonlinearity into the network by applying an activation function to the output of the previous layer. This is crucial for the network to learn complex patterns. Common activation functions, such as ReLU, Tanh, and Leaky ReLU, transform the input while keeping the output size unchanged.

- **Flattening**

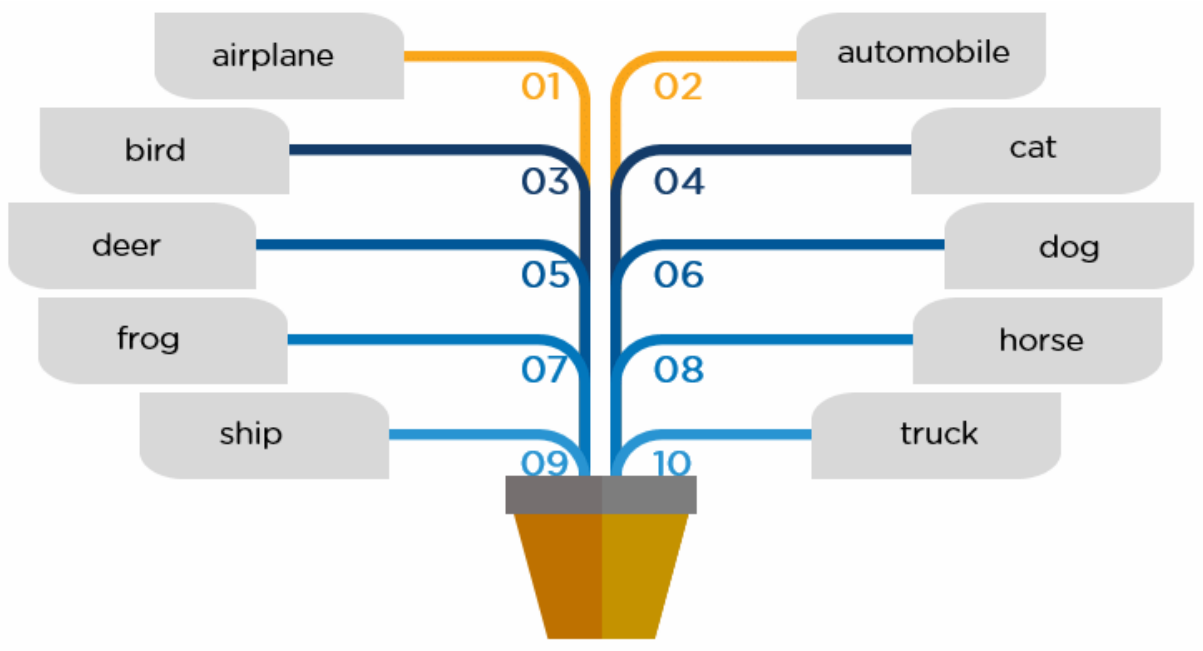
After the convolution and pooling operations, the feature maps still exist in a multi-dimensional format. Flattening converts these feature maps into a one-dimensional vector. This process is essential because it prepares the data to be passed into fully connected layers for classification or regression tasks.

- **Output Layer**

In the output layer, the final result from the fully connected layers is processed through a logistic function, such as sigmoid or softmax. These functions convert the raw scores into probability distributions, enabling the model to predict the most likely class label.

**Use Case Implementation Using CNN**

We'll use the CIFAR-10 dataset from the Canadian Institute For Advanced Research to classify images across 10 categories using CNN.




**Convolutional Neural Network Training**

Training a Convolutional Neural Network (CNN) involves guiding the model to recognize patterns in data through a step-by-step learning process. This is typically done using supervised learning, where the CNN is fed a bunch of images with their correct labels, and it gradually learns how to associate images with the right labels. Here's how the process works:

- **Data Preparation**

First things first, the images need to be prepared before training can start. This means making sure all the images are uniform in terms of format and size. By preprocessing the data in this way, you ensure that the CNN gets consistent input, which is crucial for its learning process.

- **Loss Function**

Once the images are ready, the next step is to figure out how well the CNN is doing. This is where the loss function comes into play. Think of it as a scorecard that measures the difference between what the model predicted and the actual label of the image. The smaller the difference, the better the model is performing, so the goal is to reduce this gap as much as possible.

- **Optimizer**

Now that we know how well (or poorly) the CNN is performing, it's time to improve it. The optimizer is like a coach that adjusts the network's weights to help it do better. It tweaks the model's parameters to minimize the loss function, ultimately leading to more accurate predictions over time.

- **Backpropagation**

Backpropagation is the magic behind the scenes that makes everything work. It's the process of figuring out how much each weight in the network contributed to the errors and then adjusting those weights accordingly. The optimizer uses this information to make smarter updates, helping the model get better with each round of training.

**CNN Evaluation**

Several key metrics are used to evaluate your Convolutional Neural Network after its training process is complete:

- **Accuracy**

Accuracy tells you the overall percentage of test images that the CNN correctly classifies. It's a straightforward measure of how often the model gets the right label.

- **Precision**

Precision focuses on how precise the CNN is when it predicts a particular class. It measures the percentage of test images that were predicted as a specific class and actually belong to that class. High precision means that when the CNN predicts a class, it's likely correct.

- **Recall**

Recall looks at how well the CNN identifies all instances of a particular class. It measures the percentage of test images that are of a certain class and were correctly identified as that class by the CNN. High recall indicates that the CNN is good at finding all relevant examples of a class.

- **F1 Score**

The F1 Score combines precision and recall into a single metric by calculating their harmonic mean. This is particularly useful for evaluating the CNN's performance on classes where there's an imbalance, meaning some classes are much more common than others. The F1 Score provides a balanced measure that considers both false positives and false negatives, offering a more comprehensive view of the CNN's performance.

**Types of Convolutional Neural Networks**

Here are the key types of Convolutional Neural Networks that have significantly impacted the field of image recognition:

- **LeNet**

LeNet, developed by Yann LeCun and his team in the late 1990s, is one of the earliest CNN architectures designed for handwritten digit recognition. It features a straightforward design with two convolutional and pooling layers followed by subsampling, and three fully connected layers. Despite its simplicity by today's standards, LeNet achieved high accuracy on the MNIST dataset and laid the groundwork for modern CNNs.

- **AlexNet**

AlexNet, created by Alex Krizhevsky and colleagues in 2012, revolutionized image recognition by winning the ImageNet Large Scale Visual Recognition Challenge (ILSVRC). Its architecture includes five convolutional layers and three fully connected layers, with innovations like ReLU activation and dropout. AlexNet demonstrated the power of deep learning, leading to the development of even deeper networks.

- **ResNet**

ResNet, or Residual Networks, introduced the concept of residual connections, allowing the training of very deep networks without overfitting. Its architecture uses skip connections to help gradients flow through the network effectively, making it well-suited for complex tasks like keypoint detection. ResNet has set new benchmarks in various image recognition tasks and continues to be influential.

- **GoogleNet**

GoogleNet, also known as InceptionNet, is known for its efficiency and high performance in image classification. It introduces the Inception module, which allows the network to process features at multiple scales simultaneously. With global average pooling and factorized convolutions, GoogleNet achieves impressive accuracy while using fewer parameters and computational resources.

- **MobileNet**

MobileNets are designed for mobile and embedded devices, offering a balance of high accuracy and computational efficiency. By using depth-wise separable convolutions, MobileNets reduce the model size and computational demand while maintaining strong performance in image classification and keypoint detection. Their efficiency makes them ideal for resource-constrained environments.

- **VGG**

VGG networks are recognised for their simplicity and effectiveness, using a series of convolutional and pooling layers followed by fully connected layers. Their straightforward architecture has made them popular in various image recognition tasks, including object detection in self-driving cars. VGG's design remains a powerful tool for many applications due to its versatility and ease of use.

**Applications of CNN**

Now, let's look at the various applications of CNN in machine learning and how they are used across different fields:

- **Image Classification**

CNN in deep learning excels at image classification, which involves sorting images into predefined categories. They can effectively identify whether an image depicts a cat, dog, car, or flower, making them indispensable for tasks that require sorting and labeling large volumes of visual data.

- **Object Detection**

CNNs are particularly skilled in object detection, allowing them to identify and pinpoint specific items within an image. Whether it's recognizing people, cars, or buildings, CNNs can locate these objects and highlight their positions, which is crucial for applications needing accurate object placement and identification.

- **Image Segmentation**

CNNs are highly effective for tasks that involve breaking down an image into distinct parts. Image segmentation allows CNNs to distinguish and label different objects or regions within an image. This capability is essential in fields like medical imaging, where detailed analysis of structures is required, and in robotics, where intricate scenes need to be understood.

- **Video Analysis**

CNNs are also adept at video analysis, where they can track objects and detect events over time. This makes them valuable for applications like surveillance and traffic monitoring, where continuously analyzing dynamic scenes helps in understanding and managing real-time activities.

**Advantages of CNN**

Apart from their diverse applications, here are some notable advantages of CNN in deep learning that highlight their effectiveness and versatility:

- **High Accuracy**

Convolutional Neural Networks are known for their exceptional accuracy in image recognition tasks. They perform impressively in areas like classifying images, detecting objects, and segmenting visuals, setting a high benchmark for performance in these fields.

- **Efficiency**

These networks are particularly efficient when used with specialized hardware such as GPUs. This efficiency allows CNNs to process large amounts of data quickly, which is crucial for applications that require heavy computational power.

- **Robustness**

Convolutional Neural Networks handle noisy or inconsistent input data with impressive resilience. Their ability to maintain performance despite data imperfections makes them dependable for real-world applications where conditions can vary.

- **Flexibility**

Another key advantage of Convolutional Neural Networks is their adaptability. They can be tailored to different tasks simply by altering their architecture. This makes them versatile tools that can be easily repurposed for diverse applications, from medical imaging to autonomous vehicles.

**Disadvantages of Convolutional Neural Networks (CNNs)**

Although Convolutional Neural Networks (CNNs) are powerful, they come with their own set of challenges:

- **Complexity and Training Difficulty**

CNNs are intricate, and this complexity can make them difficult to train, especially when working with large datasets. Managing and fine-tuning the layers requires a deep understanding of the architecture, making it challenging even for seasoned professionals.

- **High Computational Demands**

Another significant disadvantage is the high computational power required to train and deploy CNNs effectively. Advanced hardware, such as GPUs, is often necessary, which increases costs and limits

access for those without these resources. This makes it difficult for smaller organizations to utilize CNNs efficiently.

- **Large Data Requirements**

CNNs need a large amount of labeled data to perform well. Gathering and labeling data is time-consuming and expensive. For more complex applications, such as medical imaging, the precision needed in data labeling further increases the cost and effort involved.

- **Lack of Interpretability**

One of the most notable challenges with CNNs is their black-box nature. It's often difficult to understand why a CNN makes a certain prediction, which can be a significant issue in areas where decision-making transparency is important. This lack of interpretability can limit the trust placed in CNN-based systems, especially in critical applications like healthcare.

**Convolutional neural networks and computer vision**

CNN in machine kearning is at the heart of many computer vision applications across various industries. Here's how they're making an impact:

- **Marketing**

In marketing, social media platforms leverage CNN in machine learning to enhance user experiences. For example, platforms can suggest who might be in a posted photograph, making it easier to tag friends and share content. This adds a personal touch to social media interactions and improves engagement.

- **Healthcare**

In radiology, CNN-powered computer vision helps doctors detect cancerous tumors with greater accuracy, assisting in early diagnosis and better patient outcomes. It's revolutionizing how medical images are analyzed.

- **Retail**

E-commerce platforms use CNNs for visual search, allowing users to find products by simply uploading images. This technology also helps retailers suggest complementary items, making shopping more intuitive and engaging.

- **Automotive**

CNNs are improving automotive safety through features like lane detection and collision warnings. These advancements are bringing us closer to autonomous driving by enhancing current vehicle safety systems.

# Supervised deep learning

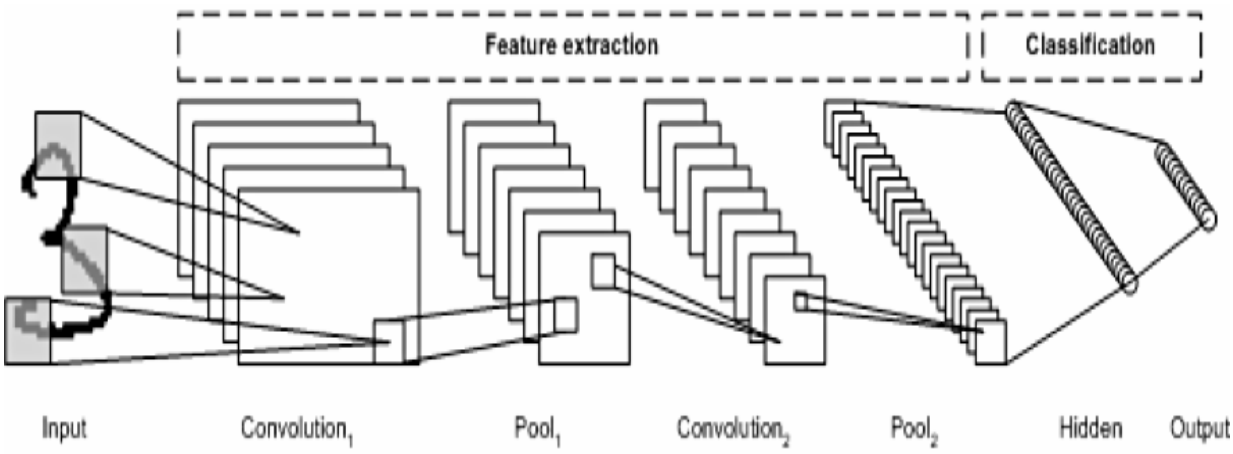
Supervised learning refers to the problem space wherein the target to be predicted is clearly labelled within the data that is used for training.

In this section, we introduce at a high-level two of the most popular supervised deep learning architectures - convolutional neural networks and recurrent neural networks as well as some of their variants.

## Convolutional neural networks

A CNN is a multilayer neural network that was biologically inspired by the animal visual cortex. The architecture is particularly useful in image-processing applications. The first CNN was created by Yann LeCun; at the time, the architecture focused on handwritten character recognition, such as postal code interpretation. As a deep network, early layers recognize features (such as edges), and later layers recombine these features into higher-level attributes of the input.

The LeNet CNN architecture is made up of several layers that implement feature extraction and then classification (see the following image). The image is divided into receptive fields that feed into a convolutional layer, which then extracts features from the input image. The next step is pooling, which reduces the dimensionality of the extracted features (through down-sampling) while retaining the most important information (typically, through max pooling). Another convolution and pooling step is then performed that feeds into a fully connected multilayer perceptron. The final output layer of this network is a set of nodes that identify features of the image (in this case, a node per identified number). You train the network by using back-propagation.



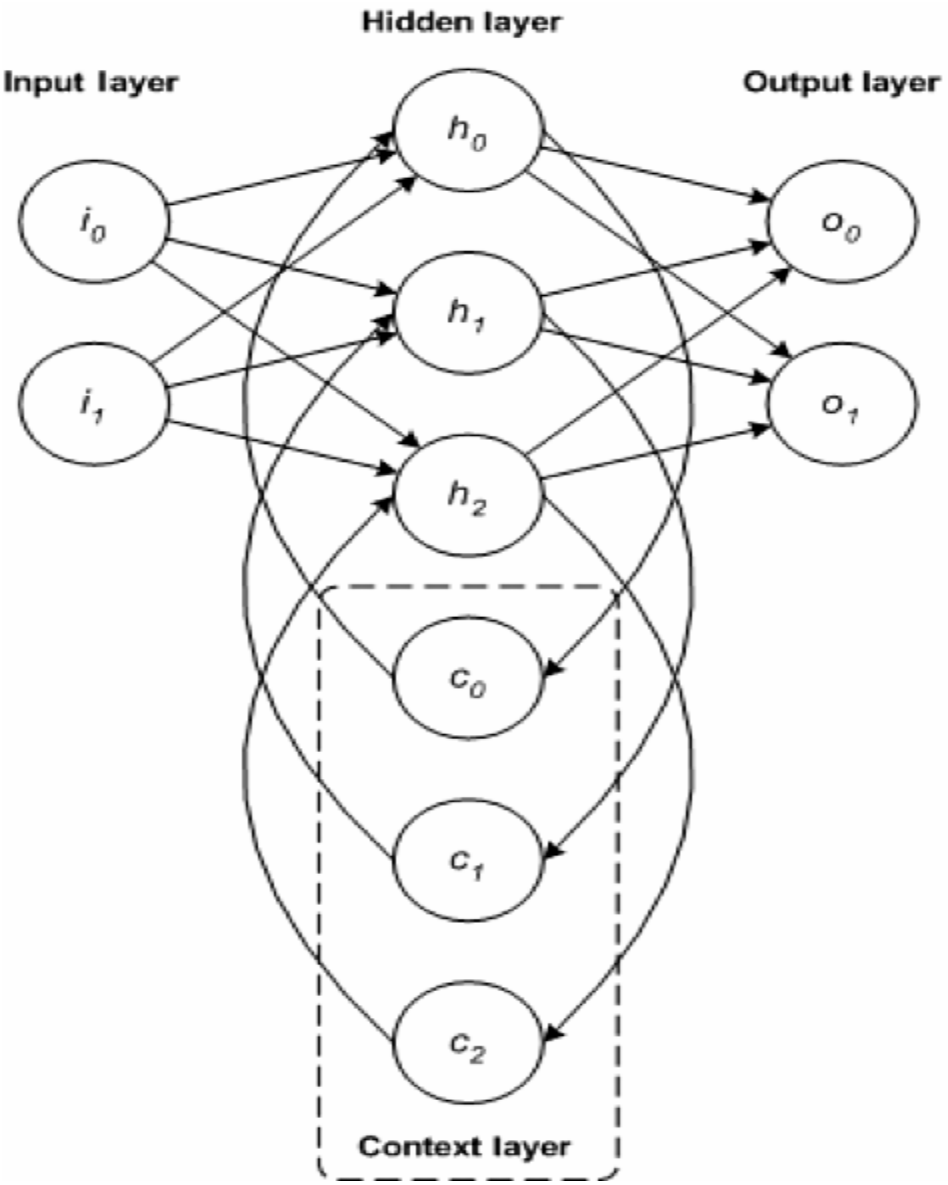

The use of deep layers of processing, convolutions, pooling, and a fully connected classification layer opened the door to various new applications of deep learning neural networks. In addition to image processing, the CNN has been successfully applied to video recognition and various tasks within natural language processing.

*Example applications: Image recognition, video analysis, and natural language processing*

## Recurrent neural networks

The RNN is one of the foundational network architectures from which other deep learning architectures are built. The primary difference between a typical multilayer network and a recurrent network is that rather than completely feed-forward connections, a recurrent network might have connections that feed back into prior layers (or into the same layer). This feedback allows RNNs to maintain memory of past inputs and model problems in time.

RNNs consist of a rich set of architectures (we'll look at one popular topology called LSTM next). The key differentiator is feedback within the network, which could manifest itself from a hidden layer, the output layer, or some combination thereof.




RNNs can be unfolded in time and trained with standard back-propagation or by using a variant of back-propagation that is called back-propagation in time (BPTT).

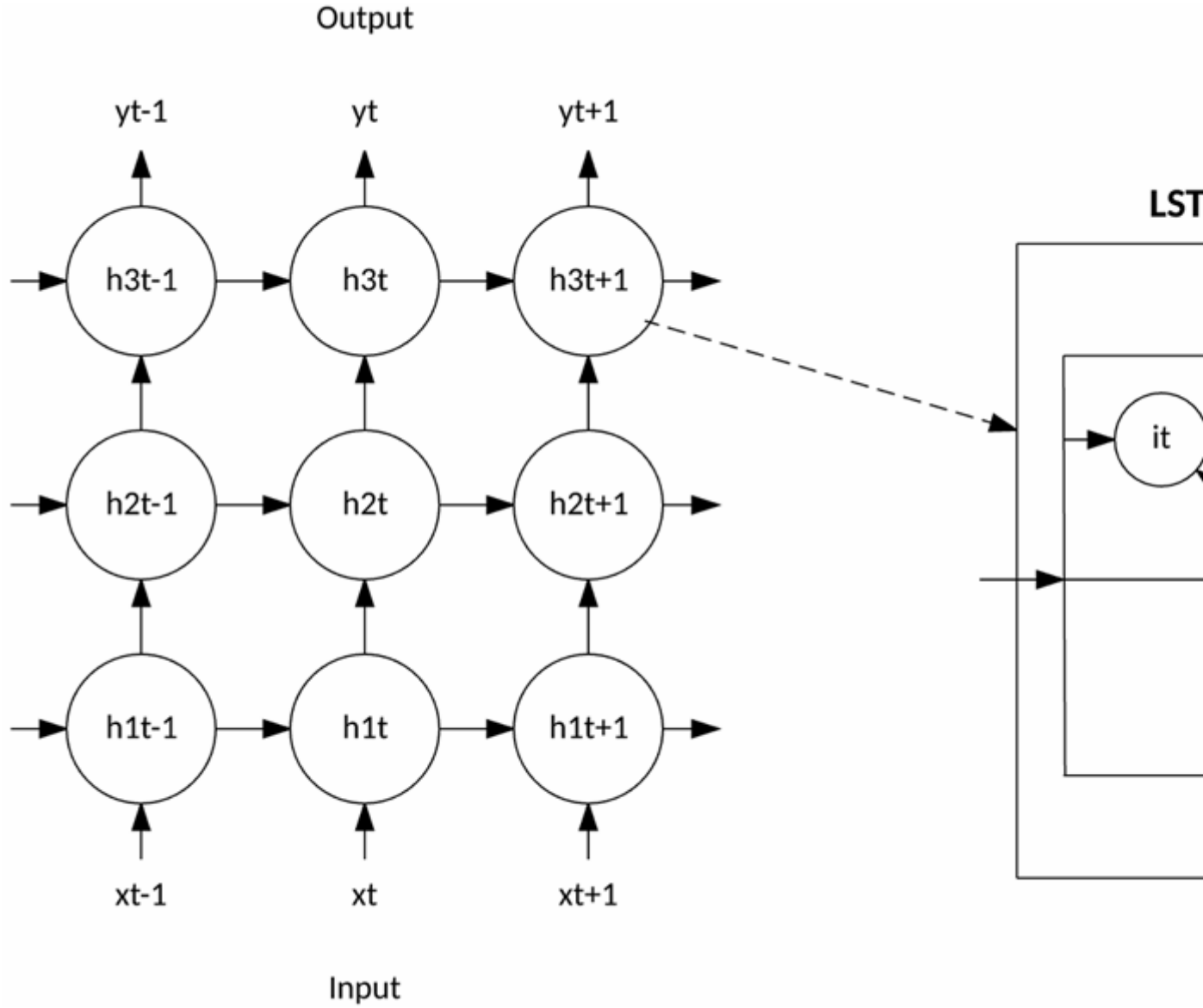
*Example applications: Speech recognition and handwriting recognition*

**LSTM networks**

The LSTM was created in 1997 by Hochreiter and Schimdhuber, but it has grown in popularity in recent years as an RNN architecture for various applications. You'll find LSTMs in products that you use every day, such as smartphones. IBM applied LSTMs in IBM Watson® for milestone-setting conversational speech recognition.

The LSTM departed from typical neuron-based neural network architectures and instead introduced the concept of a memory cell. The memory cell can retain its value for a short or long time as a function of its inputs, which allows the cell to remember what's important and not just its last computed value.

The LSTM memory cell contains three gates that control how information flows into or out of the cell. The input gate controls when new information can flow into the memory. The forget gate controls when an existing piece of information is forgotten, allowing the cell to remember new data. Finally, the output gate controls when the information that is contained in the cell is used in the output from the cell. The cell also contains weights, which control each gate. The training algorithm, commonly BPTT, optimizes these weights based on the resulting network output error.

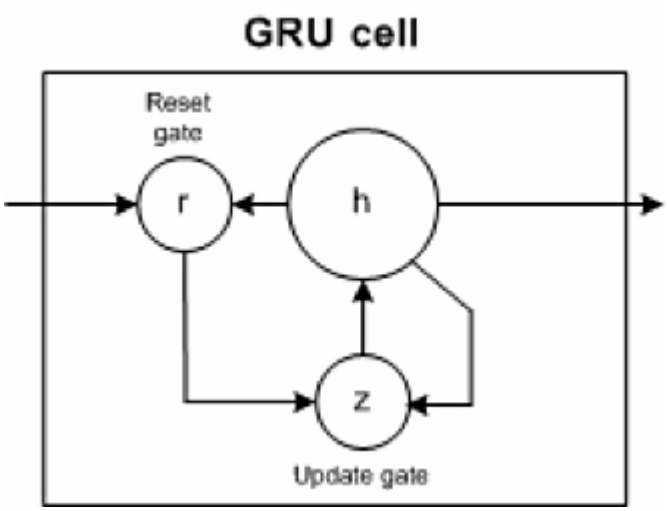



Recent applications of CNNs and LSTMs produced image and video captioning systems in which an image or video is captioned in natural language. The CNN implements the image or video processing, and the LSTM is trained to convert the CNN output into natural language.

*Example applications: Image and video captioning systems*

**GRU networks**

In 2014, a simplification of the LSTM was introduced called the gated recurrent unit. This model has two gates, getting rid of the output gate present in the LSTM model. These gates are an update gate and a reset gate. The update gate indicates how much of the previous cell contents to maintain. The reset gate defines how to incorporate the new input with the previous cell contents. A GRU can model a standard RNN simply by setting the reset gate to 1 and the update gate to 0.




The GRU is simpler than the LSTM, can be trained more quickly, and can be more efficient in its execution. However, the LSTM can be more expressive and with more data can lead to better results.

*Example applications: Natural language text compression, handwriting recognition, speech recognition, gesture recognition, image captioning*